Hanna Bae

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Education

Korea Advanced Institute of Science and Technology — *Daejeon, Korea*
M.S. in School of Electrical Engineering — *Feb 2025 - Current*
- Statistical Inference & Information Theory Laboratory advised by Professor Junmo Kim.

Gwangju Institute of Science and Technology — *Gwangju, Korea*
B.S. in Electrical Engineering and Computer Science — *Feb 2021 - Feb 2025*
- Minors: Mathematics and Artificial Intelligence
- GPA: 4.11/4.5

University of California, Berkeley — *Berkeley, CA, USA*
Summer Session — *Jun 2022 - Aug 2022*
- Data 8: Foundations of Data Science; STAT 20: Introduction to Probability and Statistics.

Research Interest

3D Vision, Geometric Perception, 3D Reconstruction, Visual SLAM, Multi-View Geometry, Autonomous Driving

Publication

Position Matters in the Overlap: Analyzing Chunk Alignment for Long-Sequence 3D Reconstruction — *Long-sequence reconstruction, Chunk alignment*
Hanna Bae, Sojeong Song, Gyojin Han, Sungjae Choi, and Junmo Kim.
Manuscript **(Under Review)**

Adaptive Dense Risk Prediction for Consistent Feed-Forward 3D Reconstruction — *3D reconstruction, Risk prediction*
Sojeong Song, **Hanna Bae**, Chang D. Yoo, and Junmo Kim.
Manuscript **(Under Review)**

Information-Aware Test-Time Update and Dynamic Token Reinitialization for Continuous 3D Reconstruction — *Continuous reconstruction, Test-time adaptation*
Sunghyun Baek, **Hanna Bae**, Minchan Kwon, and Junmo Kim.
Manuscript **(Under Review)**

VistaGS: Multi-View Geometry Foundation Priors for View-Consistent Few-Shot 3D Gaussian Splatting
Sungjae Choi, Gyojin Han, **Hanna Bae**, and Junmo Kim.
Manuscript **(Under Review)**

VGGT-Bridge: Beyond Sequential Pose Graphs via Corase-Stride Skip Edges
Sungjae Choi, **Hanna Bae**, Sunghyun Baek, and Junmo Kim.
Manuscript **(Under Review)**

Data Preprocessing Methods for Improving Korean Table Question Answering — *Table question answering, Data-centric AI*
Hanna Bae, Sanghyuk Won, Jaehong Lee, Woohyuk Jung, and Jihwan Woo.
Journal of Information Systems, 27(2), 71-87 **(2025)**

Research Experience

Autonomous Computing Systems Lab (ACSL) — *Gwangju, Korea*
Research Intern — *Mar 2023 - Jun 2024*
- Studied mobile/edge 3D reconstruction, dense correspondence, and detector-free feature matching for SLAM front ends.
- Explored deployment pipelines using TensorRT, ONNX, CoreML, and TinyML with attention to accuracy-latency-memory trade-offs.

CJ OliveNetworks AI Research Lab — *Seoul, Korea*
AI Research Intern — *Nov 2024*
- Built Korean table question-answering data with merged and composite structures and compared multiple table representations.
- Developed the internship study into a first-author journal publication.

Project Experience

Retriever-Free Visual SLAM *Daejeon, Korea*
Research Project *2025*
- Combined a MASt3R-SLAM front end with a feed-forward multi-view reconstruction back end through dense matching rather than explicit image retrieval.
- Studied keyframe selection, stationary-frame handling, loop closure, pose-graph optimization, and yaw drift in long sequences.

Camera-Radar Fusion for 3D Object Detection *Daejeon, Korea*
Research Project *2025*
- Implemented camera-radar feature fusion and teacher-student distillation; investigated radar augmentation drift, cross-modal alignment, and illumination robustness.

Detector-Free Feature Matching for SLAM Front-End *Gwangju, Korea*
B.S. Thesis *2024*
- Studied detector-free local feature matching and semantic filtering for visual SLAM and evaluated the method on MegaDepth.

OpenVINS-MASt3R Hybrid Visual SLAM *Daejeon, Korea*
Team Project *2025*
- Analyzed visual-inertial odometry under noisy IMU and depth-derived pose corruption; explored anomaly rejection and Hampel filtering for stable estimation.

On-Device AI for Raman Spectroscopy *Daejeon, Korea*
Industry-Academic Research Project *2026*
- Designed a lightweight spectral preprocessing, classification, quality assessment, and out-of-distribution detection pipeline under device-level latency constraints.

Additional Machine Learning Projects *Gwangju, Korea*
Course and Independent Projects *2023 - 2024*
- Developed sleep-stage classification, Korean text classification, profanity filtering, and an object-detection demo deployed with Streamlit.

Teaching Experience

Teaching Assistant, Differential Equations, GIST *Spring 2023; Fall 2024*
Teaching Assistant, Computer Programming, GIST *Fall 2023*
Teaching Assistant, Computer System Theory and Experiments, GIST *Spring 2024*

Skills

Programming Python (PyTorch, NumPy, pandas, OpenCV), C++
3D Vision 3D reconstruction, visual SLAM, multi-view geometry, pose estimation, dense correspondence, trajectory evaluation
Systems Linux, Git, Docker/Conda, TensorRT, ONNX, CoreML, LaTeX